python data structure

list data structure

In [1]:
```
l = []
l
```

Out[1]:
```
[]
```

In [2]:
```
type[l]
```

Out[2]:
```
type[[]]
```

In [3]:
```
print(type(l))
```

```
<class 'list'>
```

In [4]:
```
l.append(10)
```

In [5]:
```
l
```

Out[5]:
```
[10]
```

In [6]:
```
l.append(20)
l.append(30)
l.append(40)
l.append(50)
```

In [7]:
```
l
```

Out[7]:
```
[10, 20, 30, 40, 50]
```

In [8]:
```
len(l)
```

Out[8]:
```
5
```

In [9]:
```
l.append(10)
```

In [10]:
```
l
```

Out[10]:
```
[10, 20, 30, 40, 50, 10]
```

In [11]:
```
l1 = [10,2.3,1+2j,True,'nit',[1,2,3]]
```

In [12]:
```
l1
```

Out[12]:
```
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [13]:
```
len(l1)
```

Out[13]: 6

In [14]:
```
print(l)
print(l1)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [15]:
```
print(id(l))
print(id(l1))
```

```
1879021584512
1876936872896
```

In [16]:
```
l
```

Out[16]: [10, 20, 30, 40, 50, 10]

In [17]:
```
l2
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[17], line 1
----> 1 l2

NameError: name 'l2' is not defined
```

In [18]:
```
l2= l.copy()
```

In [19]:
```
l2
```

Out[19]: [10, 20, 30, 40, 50, 10]

In [20]:
```
print(id(l))
print(id(l1))
print(id(l2))
```

```
1879021584512
1876936872896
1876943026944
```

In [21]:
```
l2
```

Out[21]: [10, 20, 30, 40, 50, 10]

In [22]:
```
len(l2)
```

Out[22]: 6

In [23]:
```
l2.clear()
```

In [24]:
```
len(l2)
```

Out[24]: 0

In [25]:
```
l2
```

Out[25]: []

In [26]:
```
del l2
```

In [27]:
```
l2
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[27], line 1
----> 1 l2

NameError: name 'l2' is not defined
```

In [28]:
```
l
```

Out[28]: [10, 20, 30, 40, 50, 10]

In [29]:
```
l.count(20)
```

Out[29]: 1

In [30]:
```
l.count(30)
```

Out[30]: 1

In [31]:
```
l.count(10)
```

Out[31]: 2

In [32]:
```
l.count(100)
```

Out[32]: 0

In [33]:
```
print(l)
print(l1)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [34]:
```
a = [1,2,3]
b = [4,5,6]
a.extend(b)
```

In [35]:
```
a
```

Out[35]: [1, 2, 3, 4, 5, 6]

In [36]:
```
l2 = []
l2
```

Out[36]: []

In [37]:
```
l2.extend(l)
```

In [38]:
```
l2
```

Out[38]: [10, 20, 30, 40, 50, 10]

In [39]:
```
l2.extend(l1)
```

In [40]:
```
l2
```

Out[40]: [10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]

In [41]:
```
print(l)
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
[10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [42]:
```
l
```

Out[42]: [10, 20, 30, 40, 50, 10]

In [43]:
```
l[0]
```

Out[43]: 10

In [44]:
```
l[0] = 100
```

In [45]:
```
l
```

Out[45]: [100, 20, 30, 40, 50, 10]

In [46]:
```
job = [5000]
job
```

Out[46]: [5000]

In [47]:
```
job[0] = 10000
```

In [48]:
```
job
```

Out[48]: [10000]

In [49]:
```
l
```

Out[49]: [100, 20, 30, 40, 50, 10]

In [50]:
```
l[-2] = 500
l
```

Out[50]:
```
[100, 20, 30, 40, 500, 10]
```

In [51]:
```
l.index()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[51], line 1
----> 1 l.index()

TypeError: index expected at least 1 argument, got 0
```

In [52]:
```
l.index(100)
```

Out[52]:
```
0
```

In [53]:
```
l2
```

Out[53]:
```
[10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [54]:
```
l2.index(10)
```

Out[54]:
```
0
```

In [55]:
```
l
```

Out[55]:
```
[100, 20, 30, 40, 500, 10]
```

In [56]:
```
l.insert(2 , 25)
l
```

Out[56]:
```
[100, 20, 25, 30, 40, 500, 10]
```

In [57]:
```
l
```

Out[57]:
```
[100, 20, 25, 30, 40, 500, 10]
```

In [58]:
```
l.pop()
```

Out[58]:
```
10
```

In [59]:
```
l2
```

Out[59]:
```
[10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [60]:
```
l2.pop()
```

Out[60]:
```
[1, 2, 3]
```

In [61]:
```
l2.pop()
```

Out[61]: 'nit'

In [62]:
```
l2.pop()
```

Out[62]: True

In [63]:
```
l2.remove(40)
```

In [64]:
```
l2
```

Out[64]: [10, 20, 30, 50, 10, 10, 2.3, (1+2j)]

In [65]:
```
l
```

Out[65]: [100, 20, 25, 30, 40, 500]

In [66]:
```
l.reverse()
```

In [67]:
```
l
```

Out[67]: [500, 40, 30, 25, 20, 100]

In [68]:
```
l.sort()  # ascending order
```

In [69]:
```
l
```

Out[69]: [20, 25, 30, 40, 100, 500]

In [70]:
```
l.sort(reverse = False)
l
```

Out[70]: [20, 25, 30, 40, 100, 500]

In [71]:
```
l3 = [10,20,[2 ,3],3]
```

In [72]:
```
l3[2]=[2]
```

In [73]:
```
l3
```

Out[73]: [10, 20, [2], 3]

In [74]:
```
l
```

Out[74]: [20, 25, 30, 40, 100, 500]

In [75]:
```
30 in l
```

Out[75]: True

In [76]:
```
25 in l
```

Out[76]: True

In [77]:
```
l
```

Out[77]: [20, 25, 30, 40, 100, 500]

In [78]:
```
for i in l:
    print(i)
```

```
20
25
30
40
100
500
```

In [7]:
```
l1 = [10,2.3,1+2j,True,'nit',[1,2,3]]
```

In [8]:
```
l1[:]
```

Out[8]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]

In [9]:
```
l1[::-2]
```

Out[9]: [[1, 2, 3], True, 2.3]

In [10]:
```
l1[0:10:3]
```

Out[10]: [10, True]

In [11]:
```
l1
```

Out[11]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]

In [12]:
```
l1[2] = 45
l1
```

Out[12]: [10, 2.3, 45, True, 'nit', [1, 2, 3]]

tuple or tuple

In [13]:
```
t = ()
t
```

Out[13]: ()

In [14]:
```
type(t)
```

Out[14]: tuple

In [15]:
```
t1 = (10,20,30)
t1
```

Out[15]:
```
(10, 20, 30)
```

In [16]:
```
t1.count(10)
```

Out[16]:
```
1
```

In [17]:
```
t1.count(100)
```

Out[17]:
```
0
```

In [18]:
```
t1.index(10)
```

Out[18]:
```
0
```

In [19]:
```
t1[0]
```

Out[19]:
```
10
```

In [20]:
```
t1[0] = 100
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[20], line 1
----> 1 t1[0] = 100

TypeError: 'tuple' object does not support item assignment
```

In [21]:
```
SBI = (1234, 'cizps')
SBI
```

Out[21]:
```
(1234, 'cizps')
```

In [22]:
```
SBI[0] = 456
SBI
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[22], line 1
----> 1 SBI[0] = 456
      2 SBI

TypeError: 'tuple' object does not support item assignment
```

In [23]:
```
t1
```

Out[23]:
```
(10, 20, 30)
```

In [24]:
```
t1*3
```

Out[24]:
```
(10, 20, 30, 10, 20, 30, 10, 20, 30)
```

In [25]:
```
t1
```

Out[25]:
```
(10, 20, 30)
```

In [26]:
```
t1.remove()
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[26], line 1
----> 1 t1.remove()

AttributeError: 'tuple' object has no attribute 'remove'
```

set

In [27]:
```
s = {}
s
```

Out[27]:
```
{}
```

In [28]:
```
type(s)
```

Out[28]:
```
dict
```

In [29]:
```
s1 = set()
s1
```

Out[29]:
```
set()
```

In [30]:
```
s1 = {20, 30, 90, 10}
```

In [31]:
```
s1
```

Out[31]:
```
{10, 20, 30, 90}
```

In [32]:
```
type(s1)
```

Out[32]:
```
set
```

In [33]:
```
s1
```

Out[33]:
```
{10, 20, 30, 90}
```

In [34]:
```
s1.add(100)
```

In [35]:
```
s1
```

Out[35]:
```
{10, 20, 30, 90, 100}
```

In [36]:
```
s1.add(15)
```

In [37]:
```
s1
```

Out[37]:
```
{10, 15, 20, 30, 90, 100}
```

In [38]:
```
s1.add(5)
```

In [39]:
```
s1
```

Out[39]:
```
{5, 10, 15, 20, 30, 90, 100}
```

In [40]:
```
s2 = {5, 5.5, 'nit', 1+2j, True}
s2
```

Out[40]:
```
{(1+2j), 5, 5.5, True, 'nit'}
```

In [41]:
```
s2.add(100)
```

In [42]:
```
s2
```

Out[42]:
```
{(1+2j), 100, 5, 5.5, True, 'nit'}
```

In [43]:
```
print(s1)
print(s2)
```

```
{100, 5, 10, 15, 20, 90, 30}
{True, 100, 5.5, 5, (1+2j), 'nit'}
```

In [44]:
```
s2.clear
```

Out[44]:
```
<function set.clear()>
```

In [45]:
```
s3 = s2.copy()
```

In [46]:
```
s3
```

Out[46]:
```
{(1+2j), 100, 5, 5.5, True, 'nit'}
```

In [47]:
```
s3{0}
```

```
  Cell In[47], line 1
    s3{0}
      ^
SyntaxError: invalid syntax
```

In [48]:
```
s3{:}
```

```
  Cell In[48], line 1
    s3{:}
      ^
SyntaxError: invalid syntax
```

In [49]:
```
s3.pop()
```

Out[49]: True

In [50]:
```
s3
```

Out[50]: {(1+2j), 100, 5, 5.5, 'nit'}

In [51]:
```
s3.pop(3)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[51], line 1
----> 1 s3.pop(3)

TypeError: set.pop() takes no arguments (1 given)
```

In [52]:
```
s2
```

Out[52]: {(1+2j), 100, 5, 5.5, True, 'nit'}

In [53]:
```
s1
```

Out[53]: {5, 10, 15, 20, 30, 90, 100}

In [54]:
```
s3
```

Out[54]: {(1+2j), 100, 5, 5.5, 'nit'}

In [55]:
```
s2
```

Out[55]: {(1+2j), 100, 5, 5.5, True, 'nit'}

In [56]:
```
s2.discard(1+2j)
```

In [57]:
```
s2
```

Out[57]: {100, 5, 5.5, True, 'nit'}

In [58]:
```
s2.discard(1000)
```

In [59]:
```
s2
```

Out[59]: {100, 5, 5.5, True, 'nit'}

In [60]:
```
s2.remove(1000)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[60], line 1
----> 1 s2.remove(1000)

KeyError: 1000
```

In [61]:
```
s2
```

Out[61]: {100, 5, 5.5, True, 'nit'}

In [62]:
```
s2.add(100)
```

In [63]:
```
s2
```

Out[63]: {100, 5, 5.5, True, 'nit'}

set operation

In [64]:
```
s7 = {1,2,3,4,5}
s8 = {4,5,6,7,8}
s9 = {8,9,10}
```

In [65]:
```
s7.union(s8)
```

Out[65]: {1, 2, 3, 4, 5, 6, 7, 8}

In [66]:
```
s8.union(s9)
```

Out[66]: {4, 5, 6, 7, 8, 9, 10}

In [67]:
```
s7.union(s9)
```

Out[67]: {1, 2, 3, 4, 5, 8, 9, 10}

In [68]:
```
s7.union(s8,s9)
```

Out[68]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In [69]:
```
s8|s9
```

Out[69]: {4, 5, 6, 7, 8, 9, 10}

In [70]:
```
s8|s9|s7
```

Out[70]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In [71]:
```
print(s7)
print(s8)
print(s9)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [4]:
```
s7 = {1,2,3,4,5}
s8 = {4,5,6,7,8}
s9 = {8,9,10}
```

In [5]:
```
s7
```
Out[5]:
```
{1, 2, 3, 4, 5}
```

In [6]:
```
s7.intersection(s8)
```
Out[6]:
```
{4, 5}
```

In [7]:
```
s7.intersection(s9)
```
Out[7]:
```
set()
```

In [8]:
```
s7&s8
```
Out[8]:
```
{4, 5}
```

In [9]:
```
s8&s9
```
Out[9]:
```
{8}
```

In [10]:
```
s7
s8
s9
```
Out[10]:
```
{8, 9, 10}
```

In [11]:
```
s7.difference(s8)
```
Out[11]:
```
{1, 2, 3}
```

In [12]:
```
s8-s9
```
Out[12]:
```
{4, 5, 6, 7}
```

In [13]:
```
s9-s8
```
Out[13]:
```
{9, 10}
```

In [15]:
```
s7
```
Out[15]:
```
{1, 2, 3, 4, 5}
```

In [16]:
```
s8
```
Out[16]:
```
{4, 5, 6, 7, 8}
```

In [17]:
```
s9
```
Out[17]:
```
{8, 9, 10}
```

In [18]:
```
s7.symmetric_difference(s8)
```

Out[18]: {1, 2, 3, 6, 7, 8}

In [20]:
```
s8.symmetric_difference(s9)
```

Out[20]: {4, 5, 6, 7, 9, 10}

In [21]:
```
s8^s9
```

Out[21]: {4, 5, 6, 7, 9, 10}

In [22]:
```
s7
```

Out[22]: {1, 2, 3, 4, 5}

In [23]:
```
s8
```

Out[23]: {4, 5, 6, 7, 8}

In [24]:
```
s9
```

Out[24]: {8, 9, 10}

In [25]:
```
s7.symmetric_difference_update(s8)
```

In [26]:
```
s7
```

Out[26]: {1, 2, 3, 6, 7, 8}

In [27]:
```
s8.symmetric_difference_update(s9)
```

In [28]:
```
s8
```

Out[28]: {4, 5, 6, 7, 9, 10}

superset, subset, disjoint

In [29]:
```
a = {1,2,3,4,5,6,7,8,9}
b = {3,4,5,6,7,8}
c = {10,20,30,40}
```

In [30]:
```
a.issuperset(b)
```

Out[30]: True

In [31]:
```
a.issuperset(c)
```

Out[31]: False

In [32]:
```
b.issubset(a)
```

Out[32]: True

In [33]:
```
c.issubset(a)
```

Out[33]: False

In [35]:
```
print(a)
print(b)
print(c)
```

```
{1, 2, 3, 4, 5, 6, 7, 8, 9}
{3, 4, 5, 6, 7, 8}
{40, 10, 20, 30}
```

In [36]:
```
c.isdisjoint(a)
```

Out[36]: True

In [37]:
```
c.isdisjoint(b)
```

Out[37]: True

In [38]:
```
a1 = {1,2,3,4,5,6,7,8,9}
b1 = {30,40,50,60,70,80}
c1 = {10,20,30,40}
```

In [39]:
```
a1.issuperset(b1)
```

Out[39]: False

In [40]:
```
a1.issubset(c1)
```

Out[40]: False

In [41]:
```
a1.isdisjoint(b1)
```

Out[41]: True

dictionary

In [42]:
```
s = {10,20,30}
```

In [43]:
```
type(s)
```

Out[43]: set

In [44]:
```
d = {1:'one', 2:'two', 3:'three'}
```

In [45]:
```
type(d)
```

Out[45]: dict

In [46]:
```
d
```

Out[46]: {1: 'one', 2: 'two', 3: 'three'}

In [47]:
```
d.keys()
```

Out[47]: dict_keys([1, 2, 3])

In [48]:
```
d.values()
```

Out[48]: dict_values(['one', 'two', 'three'])

In [49]:
```
d
```

Out[49]: {1: 'one', 2: 'two', 3: 'three'}

In [50]:
```
d.items()
```

Out[50]: dict_items([(1, 'one'), (2, 'two'), (3, 'three')])

In [51]:
```
d
```

Out[51]: {1: 'one', 2: 'two', 3: 'three'}

In [52]:
```
d[4] = 'four'
```

In [53]:
```
d
```

Out[53]: {1: 'one', 2: 'two', 3: 'three', 4: 'four'}

In [54]:
```
d['five'] = 5
```

In [55]:
```
d
```

Out[55]: {1: 'one', 2: 'two', 3: 'three', 4: 'four', 'five': 5}

range

In [56]:
```
range(5)      # range with 1
```

Out[56]: range(0, 5)

In [57]:
```
list(range(0,5))
```

Out[57]: [0, 1, 2, 3, 4]

In [58]:
```
for i in range(0,5):
```

```
    print(i)
```

```
0
1
2
3
4
```

In [59]:
```
range(10,20)
```

Out[59]:
```
range(10, 20)
```

In [60]:
```
list(range(20))
```

Out[60]:
```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]
```

In [61]:
```
list(range(0,20,5))
```

Out[61]:
```
[0, 5, 10, 15]
```

In [62]:
```
list(range(10,20,3))
```

Out[62]:
```
[10, 13, 16, 19]
```

In [63]:
```
range(0,1,2,10,3)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[63], line 1
----> 1 range(0,1,2,10,3)

TypeError: range expected at most 3 arguments, got 5
```

In [ ]: